Unit

32 Synthesis of carbon compounds

Unit Key Concepts

- Synthetic routes for carbon compounds
- Preparation of simple carbon compounds

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

32.1 Planning a synthesis

There are many reasons for carrying out the synthesis of carbon compounds. In the pharmaceutical industry, new carbon compounds are designed and synthesized in the hope that some useful new drugs may be developed (Fig. 32.1). In the chemical industry, syntheses are done to devise more economical routes to known compounds.



Fig. 32.1 In the pharmaceutical industry, carbon compounds are designed and synthesized in the hope that useful new drugs may be developed

Most carbon compounds are made in a series of steps. There is often no single correct solution to a synthesis problem, and several alternatives may well be equally viable.

Many factors will influence the final choice of a synthetic route. These include the following:

- The availability of a suitable starting material.
- The number of stages needed to convert the starting material to the carbon compound required.
- The percentage yield that can be expected from each of the stages.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

In a synthesis with several steps, a few steps with low yields can have a disastrous effect on the **overall yield**, and the material may efficiently disappear before the synthesis is complete.

Consider a synthesis that involves 3 steps, each of which gives a 70% yield. The overall yield after the 3 steps would be:

$$\frac{70}{100} \times \frac{70}{100} \times \frac{70}{100} = \frac{34.3}{100} \text{ or } 34.3\%$$

- Whether side reactions can occur at any of the stages of the synthesis, resulting in lower yields and contamination.
- Whether there are any additional problems involved in scaling up the synthesis to manufacturing proportions, particularly the relative costs of starting materials, special designs for the chemical plant and safety considerations.

Discussion

This discussion aims at raising students' awareness of the principles of green chemistry. See the following list of green chemistry principles the discussion questions help to convey.

1 Safer auxiliary substances — Whenever possible, avoid the use of auxiliary substances such as solvents and separating agents. When these substances are needed, they should be non-toxic.

2 Energy efficiency — Energy efficiency should be considered when designing and producing a product.

3 Safer auxiliary substances — Whenever possible, avoid the use of auxiliary substances such as solvents and separating agents. When these substances are needed, they should be non-toxic.

4 Safer auxiliary substances — Whenever possible, avoid the use of auxiliary substances such as solvents and separating agents. When these substances are needed, they should be non-toxic.

5 Use of renewable resources — When possible, the feedstocks should come from a renewable resource.

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Discuss your choice in each of the following cases when you are designing a synthetic route for a carbon compound.

1 A reaction that uses dichloromethane as a solvent, versus one that uses water as a solvent and versus one that uses no solvent.

2 A reaction that takes place at 200 °C versus one that takes place at room temperature.

3 A reaction that requires the use of a drying agent versus one that does not require a drying agent.

4 A reaction that requires purification of the product versus one that requires no purification of the product.

5 A synthesis that uses a starting material that is produced from petroleum versus one that uses a *biomass-derived* starting material.

overall yield 總百分產率　　biomass-derived 由生物量衍生的

Planning a synthetic route for a carbon compound depends on the knowledge of different reactions of carbon compounds. A summary of the reactions of carbon compounds that we have learnt would help.

Practice 32.1

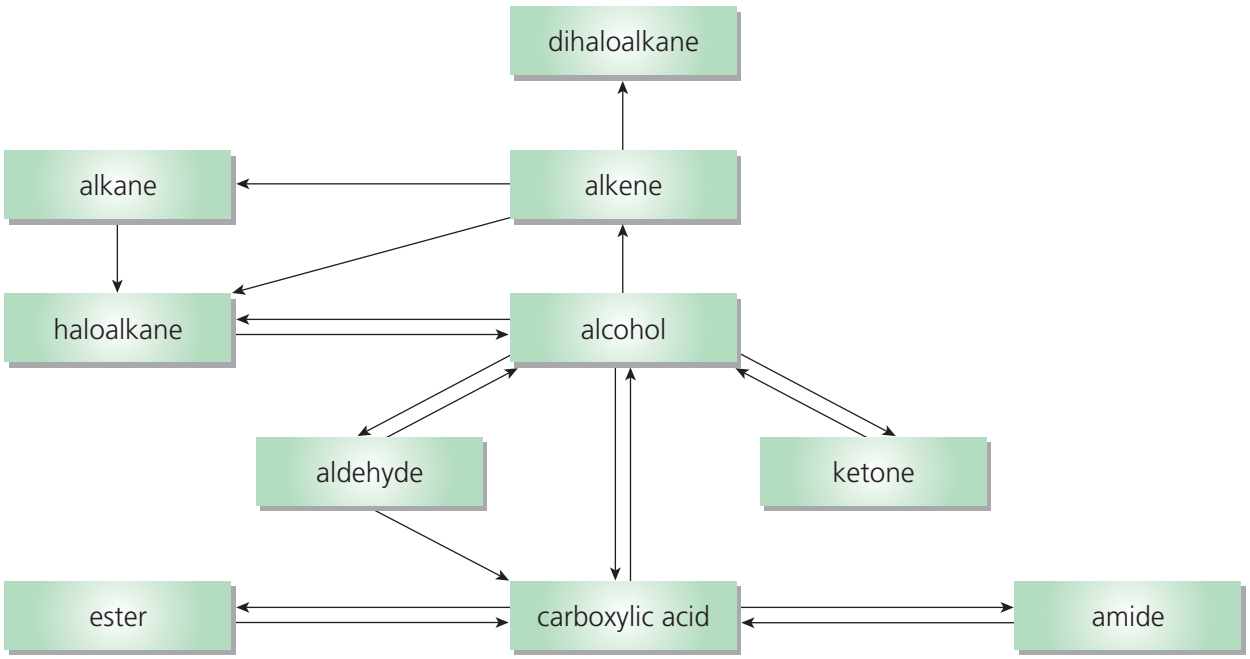
1 Prepare a flow chart which summarizes the reactions that you have learnt in carbon chemistry.
Refer to the chart on page 158 of the textbook.

a) Your flow chart should include the reactions of the following homologous series:

alkanes, alkenes, haloalkanes, alcohols, carboxylic acids and their **derivatives**

b) Include in your flow chart the typical reagent(s) and reaction condition(s) needed for each functional group conversion. Write down the reaction type (e.g. substitution, oxidation, etc.) for each case.

You may prefer to create a flow chart in your own way, but a possible framework is given below.



2 Give examples of compounds in the flow chart by using ethene as the alkene.
Refer to page T48.

derivative 衍生物

Animation
Important reactions in carbon chemistry

32.2 Two-step synthetic routes

The starting point in planning a synthesis is to examine the required compound itself. This compound is called the *target molecule*. Then we work backwards through a series of steps to find a suitable starting molecule. N1

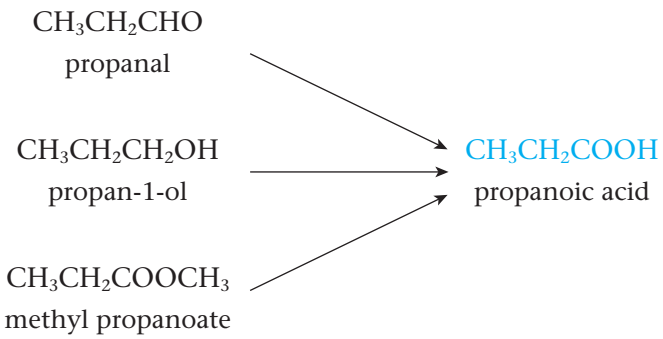
N1 The synthesis discussed would NOT involve a change in the length of the carbon chain.

For example, we need to obtain propanoic acid from 1-bromopropane.

$$\underset{\text{1-bromopropane}}{CH_3CH_2CH_2Br} \xrightarrow{?} \underset{\text{propanoic acid}}{CH_3CH_2COOH}$$

Both the starting molecule and the target molecule have the same number of carbon atoms, so no alteration to the carbon skeleton is needed.

We begin by writing down the chemical formulae of those compounds that can be converted to propanoic acid in one step. These include the aldehyde propanal, the alcohol propan-1-ol and the ester methyl propanoate.



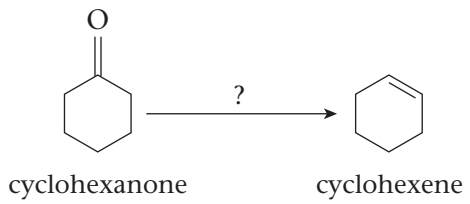
Then we look at the starting molecule, 1-bromopropane, to see whether it could be converted to one of the three compounds mentioned above.

It is obvious that we can convert 1-bromopropane into propan-1-ol by substitution with $OH^-(aq)$. So, the synthesis can be done in two steps:

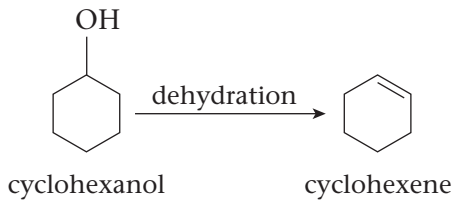
$$CH_3CH_2CH_2Br \xrightarrow[\text{reflux}]{NaOH(aq)} CH_3CH_2CH_2OH \xrightarrow[\text{reflux}]{K_2Cr_2O_7 \,/\, H_3O^+} CH_3CH_2COOH$$

target molecule 目標分子

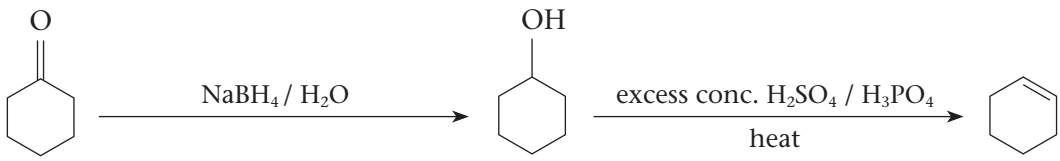
The second example is to obtain cyclohexene from cyclohexanone.



Cyclohexanol can be converted to cyclohexene by dehydration.

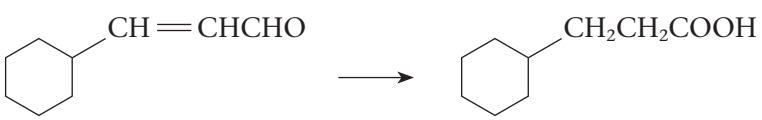


It is obvious that we can convert cyclohexanone into cyclohexanol by reduction. So, the synthesis can be done in two steps:



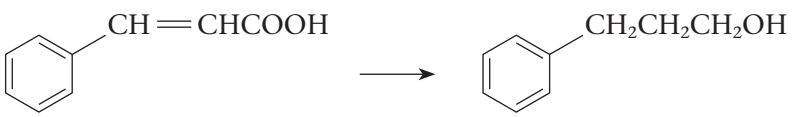
Example 32.1

Q Suggest a synthetic route for each of the following conversions.

a) 

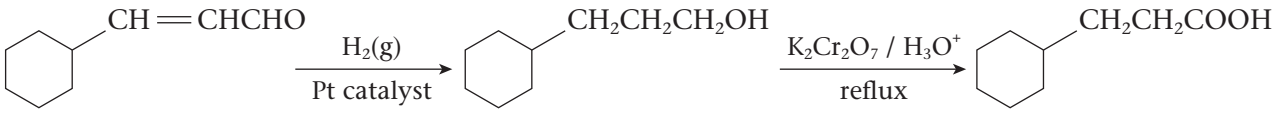
DSE 2012 Paper 1B Q12(d)

b) $CH_3CONH_2 \longrightarrow CH_3COOCH_3$

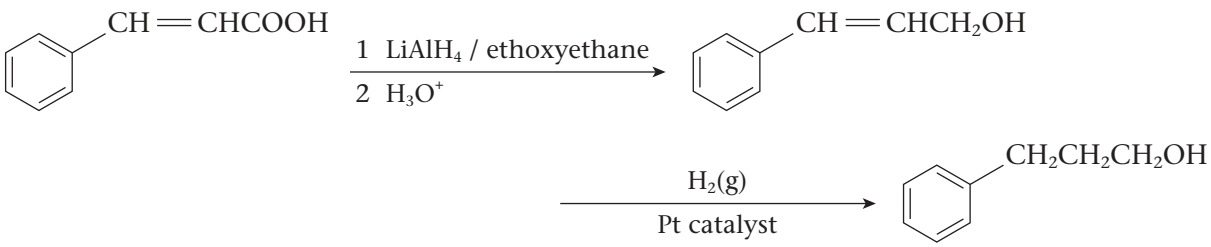
c) 

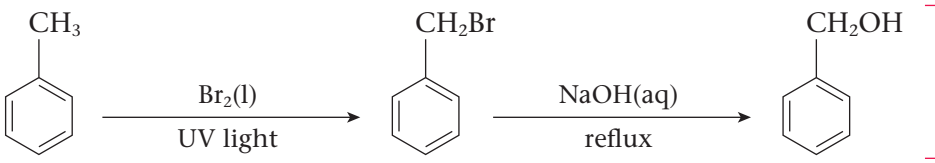
Continued on next page

d) 

A1 a) 

b) $CH_3CONH_2 \xrightarrow[\text{heat}]{\text{HCl(aq)}} CH_3COOH \xrightarrow[\text{conc. } H_2SO_4 \text{ / heat}]{CH_3OH} CH_3COOCH_3$

c) 

d) 

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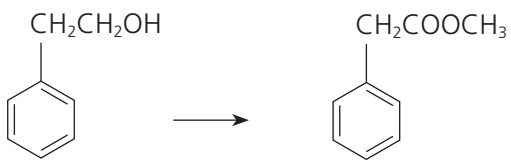
Practice 32.2

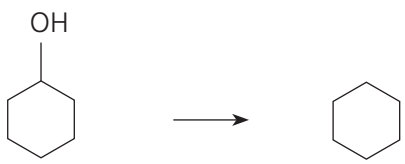
Refer to page T49.

Design simple two-step synthetic routes for the following conversions. In each case, write down your suggested route, using structural formulae. Give the reagents and reaction conditions over the arrows between compounds.

a) $CH_3CH_2COOH \longrightarrow CH_3CH_2CH_2Cl$

DSE 2014 Paper 1B Q12(b)

b) 

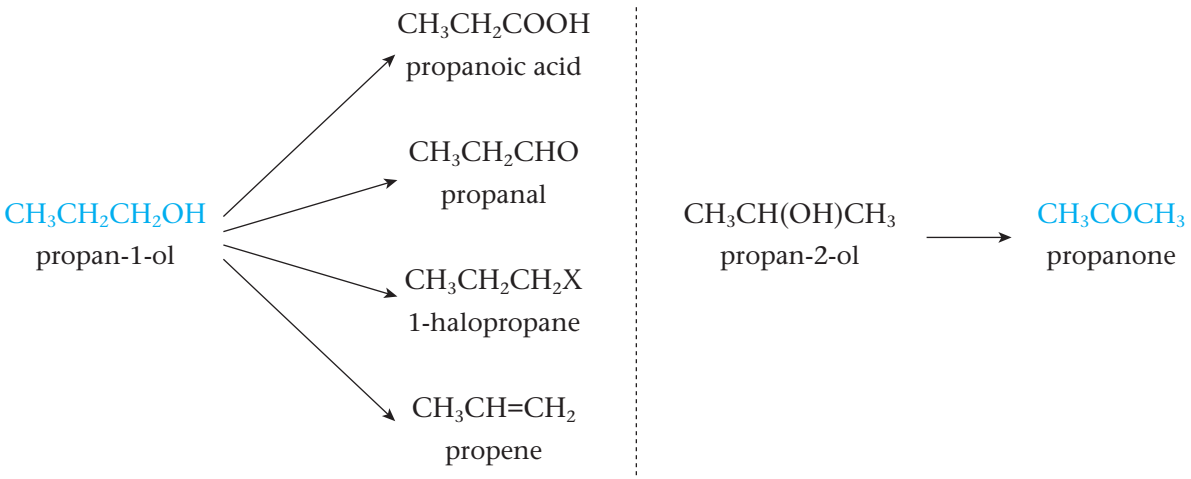
c) 

32.3 More complicated synthetic routes

Now we are going to plan synthetic routes involving more than two steps. Suppose we need to obtain propanone from propan-1-ol.

$$\underset{\text{propan-1-ol}}{CH_3CH_2CH_2OH} \xrightarrow{?} \underset{\text{propanone}}{CH_3COCH_3}$$

First write down the compounds from which propanone can be made in one step, and then compounds which can be made from propan-1-ol in one step.



A two-step synthetic route is not possible in this case. Notice that propan-2-ol, which can be converted to propanone, has an oxygen atom bonded to the second carbon atom.

Let us look at the molecules of compounds that we can make from propan-1-ol. Most of them have their functional groups on the first carbon atom. How can we get a functional group onto the second carbon atom?

We have discussed this rule in Unit 31.

The answer is via propene ($CH_3CH{=}CH_2$). Addition of HBr to it obeys the Markovnikov's rule. The major product is 2-bromopropane with the functional group (–Br) attached to the second carbon atom. We can convert 2-bromopropane to propan-2-ol. Then we can convert the propan-2-ol to propanone by oxidation.

So, the synthesis can be achieved in the following way:

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$$\underset{\text{propan-1-ol}}{CH_3CH_2CH_2OH} \xrightarrow[300\ ^\circ C]{Al_2O_3} \underset{\text{propene}}{CH_3CH{=}CH_2} \xrightarrow{HBr(g)} \underset{\text{2-bromopropane}}{CH_3CHBrCH_3}$$

$$\xrightarrow[\text{reflux}]{NaOH(aq)} \underset{\text{propan-2-ol}}{CH_3CH(OH)CH_3} \xrightarrow[\text{reflux}]{K_2Cr_2O_7 / H_3O^+} \underset{\text{propanone}}{CH_3COCH_3}$$

Chemists normally seek a synthetic route that has the least number of steps and produces a high yield of the product.

Example 32.2

Q Suggest a synthetic route for the following conversion:

$$CH_3CH_2CH_2COOH \longrightarrow CH_3CH_2CH_2CH_3$$

A

$$CH_3CH_2CH_2COOH \xrightarrow[2\ H_3O^+]{1\ LiAlH_4\ /\ \text{ethoxyethane}} CH_3CH_2CH_2CH_2OH$$

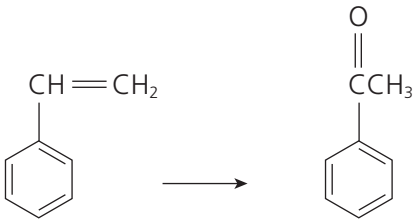
$$\xrightarrow[180\ ^\circ C]{\text{excess conc. } H_2SO_4} CH_3CH_2CH{=}CH_2 \xrightarrow[\text{Pt catalyst}]{H_2} CH_3CH_2CH_2CH_3$$

Practice 32.3

Refer to page T49.

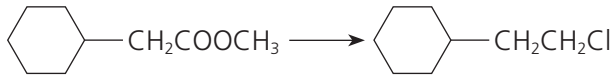
Design the following conversions. In each case, write down your suggested route, using structural formulae. Give the reaction conditions over the arrows between compounds.

a)



(Remark: assume that the benzene ring is inert to oxidizing agents.)

b) $CH_3CH_2CH_2Br \longrightarrow CH_3CH_2CONH_2$

c) 

An important part of the planning stage is a risk assessment. We will discuss this in Topic 15 Analytical Chemistry.

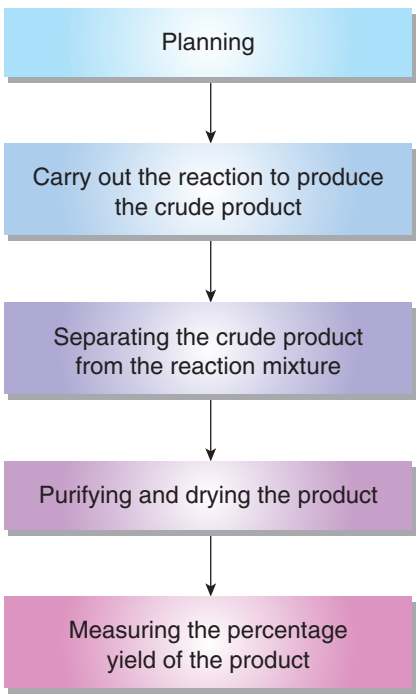


Fig. 32.2 Key stages in the preparation of a carbon compound

32.4 Laboratory preparation of simple carbon compounds

There are five key stages in the preparation of a carbon compound (Fig. 32.2).

Stage 1 Planning

The starting point of any synthesis is to choose an appropriate reaction or series of reactions. The next thing to do is to work out suitable reacting quantities from the equation and decide on the conditions for the reaction(s).

Stage 2 Carrying out the reaction to produce the crude product

Most reactions involving carbon compounds are slow at room temperature so it is usually necessary to heat the reactants. One common technique is to heat the reaction mixture under reflux.

Carbon compounds do not usually mix with aqueous reagents. Thus, another common technique is to shake the immiscible reactants in a separating funnel.

Stage 3 Separating the crude product from the reaction mixture

The result of any carbon compound preparation is a mixture of the crude product, unreacted reagents and by-products. We need to separate the crude product from the mixture.

Stage 4 Purifying and drying the product

Stage 5 Measuring the percentage yield of the product

Usually we also check the purity of the product in the final stage. We will discuss how to check that in Topic 15 Analytical Chemistry.

Reactions involving carbon compounds rarely go to completion and there are usually competing side reactions. The yield is often as low as 60%. Purification processes also lower the yield.

32.5 Common separation and purification methods in carbon compound preparations

Table 32.1 lists the common separation and purification methods we usually employ for a liquid product and a solid product in carbon compound preparations.

Table 32.1

Common separation and purification methods in carbon compound preparations

Type of product	Separation and purification method to employ
Liquid product	• distillation • fractional distillation • liquid-liquid extraction
Solid product	• re-crystallization

Distillation

We can separate a mixture of miscible liquids by **distillation** if the boiling points of the liquids are sufficiently different (say a difference of 40 °C to 50 °C) (Fig. 32.3). By heating the mixture, each pure component is vaporized, re-condensed and collected. The components will distil off in the order of their boiling points — the one with the lowest boiling point will distil off first.

For the distillation of low boiling flammable liquids, we usually use a water bath for heating. Otherwise, we may use an oil bath or a Bunsen flame.

The anti-bumping granules can ensure even boiling.

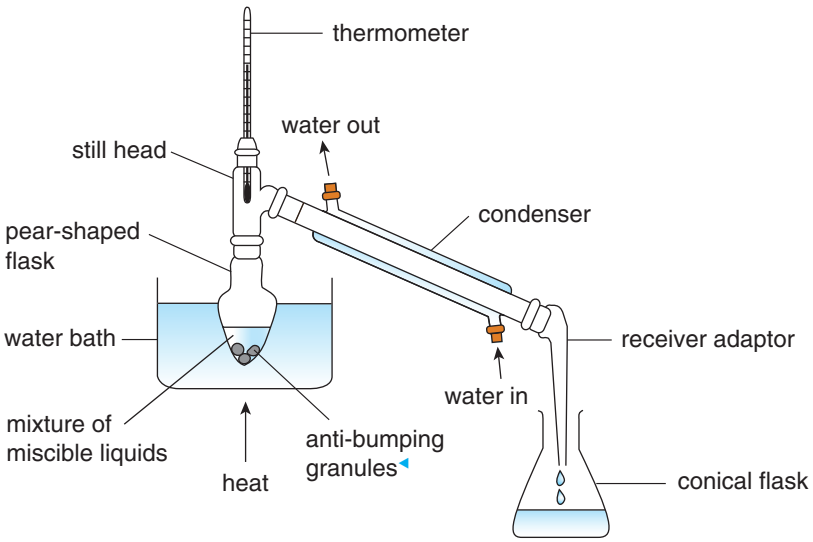




Fig. 32.3 A distillation set-up using a water bath

Fractional distillation

If we distil a mixture of two miscible liquids (such as ethanol and water) which do not have greatly different boiling points (78 °C and 100 °C), then the distillate is not a pure substance. Instead, it is a mixture that is richer in the lower boiling point component. We can separate a mixture of miscible liquids with fairly close boiling points by **fractional distillation**.

Fractional distillation is used in industry to separate both oxygen and nitrogen from liquid air. It is also used to increase the concentration of ethanol in whisky production.

distillation 蒸餾法　　fractional distillation 分餾法

To separate a mixture of miscible liquids by fractional distillation, use the apparatus shown in Fig. 32.4. The **fractionating column** is packed with glass beads. This provides a larger surface area for vaporization and condensation of the liquid mixture.

AL 2011 Paper 1 Q8(e)

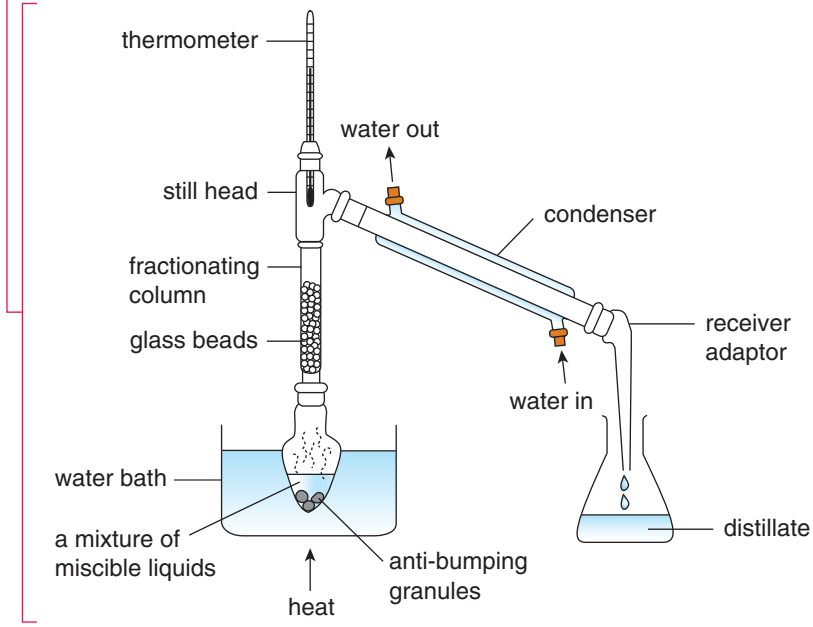




Fig. 32.4 Experimental set-up for the fractional distillation of a mixture of miscible liquids

When we heat a mixture of liquids, the vapour which comes off is a mixture of substances with different boiling points.

The column is hotter at the bottom and cooler at the top. Most of the rising vapour condenses in the column and runs back. As it does so, it meets more of the rising vapour. Some of the condensed liquid vaporizes again. In this way, the mixture vaporizes and condenses repeatedly as it rises up the column. In doing so, the rising vapour consists more and more of the lowest boiling component. At the top of the column, the vapour consists mainly of the lowest boiling component. So, the most volatile component with the lowest boiling point distils off first, then the component with the next lowest boiling point and so on.

In general, for fractional distillation to work best, the difference in boiling points of liquids in the mixture should be greater than 10 °C. Otherwise, the separation is incomplete.

fractionating column 分餾柱

Liquid-liquid extraction

In the preparation of carbon compounds, it frequently happens that the liquid product is obtained in an aqueous solution together with ionic impurities. We can use the **liquid-liquid extraction** method to extract the product.

During the extraction, place the aqueous solution containing the product in a **separating funnel**. Then add an organic solvent that is immiscible with water, such as ethoxyethane◂ (Fig. 32.5).

ASL 2009 Paper 1 Q7(e)
AL 2009 Paper 1 Q8(e)

◂ Ethoxyethane is highly flammable. Carry out the extraction inside a fume cupboard.

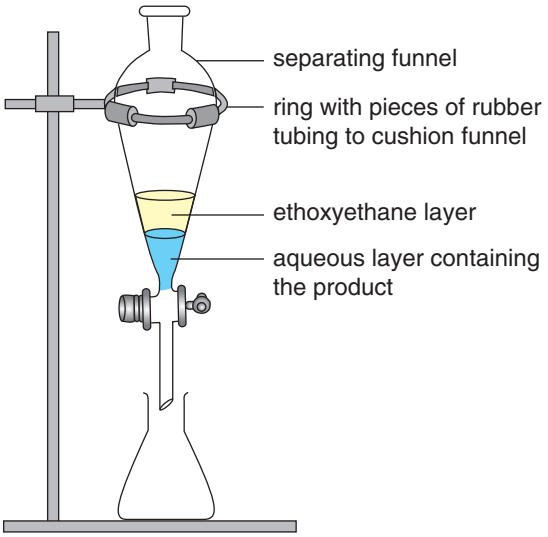




Fig. 32.5 During the extraction, add ethoxyethane to the aqueous solution containing the carbon compound product

Upon shaking◂ the liquid mixture in the separating funnel, the product will dissolve preferentially in the organic solvent. Hold the stopper and the tap firmly when shaking the separating funnel◂ (Fig. 32.6).

◂ Shaking allows a greater area of contact between the two immiscible solvents, resulting in a faster distribution of the carbon compound product between the two layers.

◂ The boiling point of ethoxyethane is 35 °C. During shaking, open the tap of the separating funnel regularly to release the pressure due to the ethoxyethane vapour caused by shaking and the heat of the hands.



Fig. 32.6 Hold the stopper and the tap firmly when shaking the separating funnel

liquid-liquid extraction 液液萃取法　　separating funnel 分液漏斗

Remove the stopper first when running off the lower layer from the separating funnel. The liquid will not run off without releasing the pressure.

Use a water bath when distilling off the ethoxyethane; use an ice-cooled receiver flask to receive the ethoxyethane.

After shaking, allow the two layers to separate and run off the lower aqueous layer. Remove the final traces of water in ethoxyethane using a drying agent such as anhydrous sodium sulphate. Filter off the drying agent. Obtain the liquid product by distilling off the ethoxyethane solvent.

ASL 2009 Paper 1 Q7(e)
AL 2009 Paper 1 Q8(e)

Re-crystallization

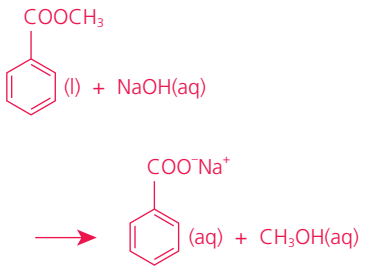
The **re-crystallization** method is often used to purify crude solid products. We use this technique when the solid product dissolves readily in a hot solvent but is much less soluble when the solvent cools down.

A re-crystallization process involves three main stages:

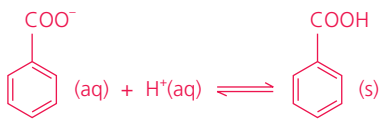
Stage 1	Choose a suitable solvent which would not react with the solid product.
Stage 2	Dissolve the crude solid product in the minimum amount of hot solvent. Filter to remove the insoluble impurities.
Stage 3	Allow the filtrate to cool slowly for crystals to re-form while the soluble impurities stay in the cold solvent. Filter off the crystals. Wash and dry them.

N2 Preparation of benzoic acid

Preparation of benzoic acid (C_6H_5COOH) involves heating methyl benzoate ($C_6H_5COOCH_3$) with excess sodium hydroxide solution under reflux for some time. The resultant mixture contains sodium benzoate and methanol.



Obtain crude benzoic acid from the resultant mixture by adding excess dilute sulphuric acid / hydrochloric acid.



The crude sample of benzoic acid can be purified by re-crystallization from water.

- Use the minimum amount of hot water to dissolve the crude solid product. Filter to remove the insoluble impurities.
- Allow the filtrate to cool slowly for crystals to re-form. Collect the crystals by filtration and dry them.

32.6 Preparing 1-bromobutane in the laboratory N2

We can prepare 1-bromobutane by the reaction of butan-1-ol with a mixture of sodium bromide and concentrated sulphuric acid.

The sodium bromide and concentrated sulphuric acid react to produce hydrogen bromide. Hydrogen bromide reacts with butan-1-ol to form 1-bromobutane.

$$NaBr(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HBr(g)$$

$$CH_3(CH_2)_2CH_2OH(aq) + HBr(g) \longrightarrow CH_3(CH_2)_2CH_2Br(l) + H_2O(l)$$

re-crystallization 再結晶法

Caution: Butan-1-ol is highly flammable. Its vapour is harmful.

Caution: Concentrated sulphuric acid is highly corrosive. It gives severe burns to the skin. Handle it with great care.

Always add concentrated sulphuric acid to water and NEVER do it the other way round.

Caution: 1-Bromobutane is highly flammable. Its vapour is irritating.

After deciding the reagents for the preparation of 1-bromobutane, the laboratory work involves four main stages:

- carrying out the reaction of butan-1-ol with a mixture of sodium bromide and concentrated sulphuric acid;
- separating the crude product from the reaction mixture;
- purifying and drying the product; and
- calculating the percentage yield of the product.

Reaction of butan-1-ol with a mixture of sodium bromide and concentrated sulphuric acid

Step 1 Heat a mixture of butan-1-ol, sodium bromide, water and concentrated sulphuric acid under reflux for about 30–45 minutes (Fig. 32.7). The flask now contains crude 1-bromobutane.

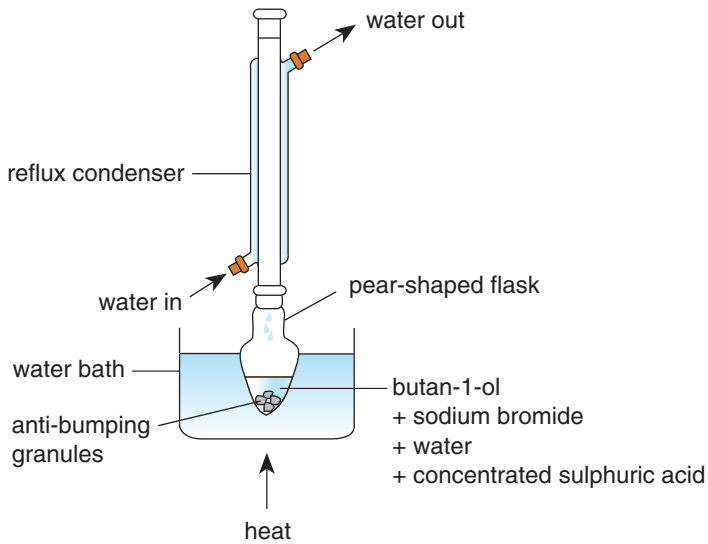




Fig. 32.7 An experimental set-up for heating the reaction mixture under reflux

Separating the crude product from the reaction mixture

Step 2 Rearrange the apparatus for distillation as shown in Fig. 32.8. Heat the mixture. 1-bromobutane and unreacted butan-1-ol are distilled off together with some acidic impurities.

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AL 2011 Paper 1 Q8(e)

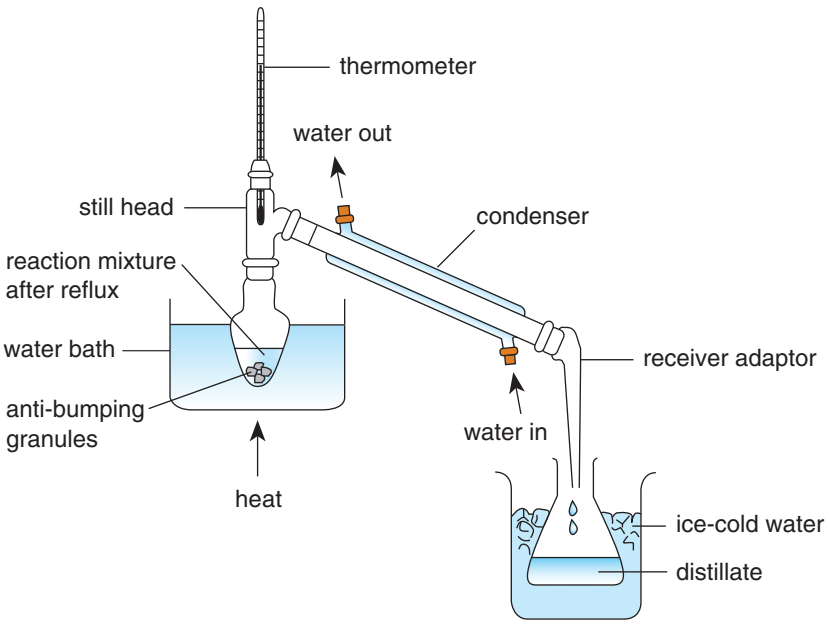


Fig. 32.8 An experimental set-up for distilling the crude product from the reaction mixture

Purifying and drying the product

Step 3 The distillate collected has a lower organic and an upper aqueous layer◀. 1-bromobutane is in the organic layer. Use a *teat pipette* to remove the aqueous upper layer (Fig. 32.9).

◀ The density of 1-bromobutane is 1.3 g cm^{-3} while that of water is 1.0 g cm^{-3}.

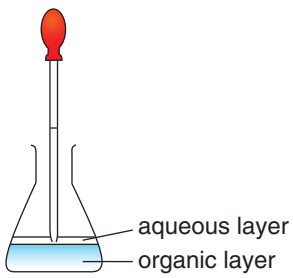


Fig. 32.9 Using a teat pipette to remove the upper aqueous layer

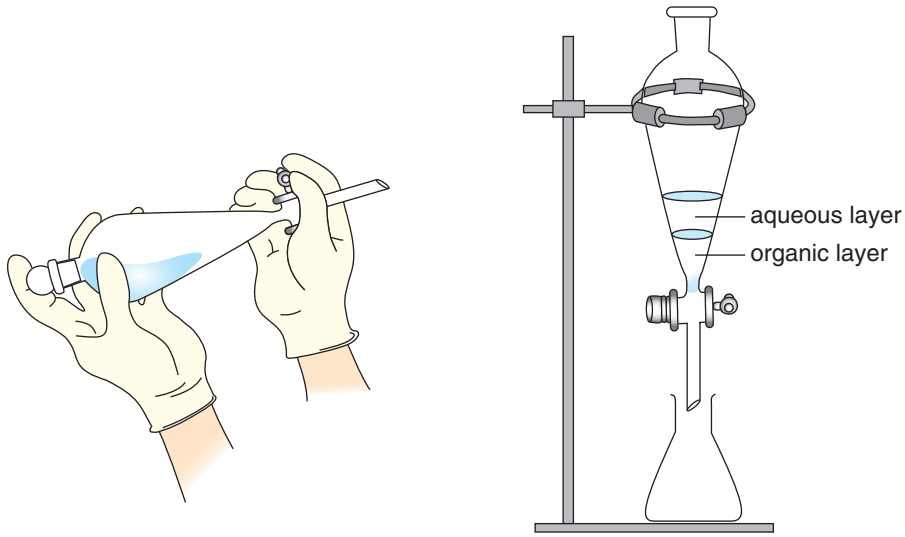
teat pipette 橡皮頭移液管

Caution: Concentrated hydrochloric acid is highly corrosive. It gives severe burns to the skin. Handle it with great care.

Step 4 Transfer the organic layer to a separating funnel and add concentrated hydrochloric acid. Stopper and shake well (Fig. 32.10a). This removes the unreacted butan-1-ol as it is soluble in the acid.

Allow the two layers to separate (Fig. 32.10b). Run the organic layer into a conical flask.

During shaking, open the tap of the separating funnel regularly to release the pressure.



(a) Shaking the separating funnel (b) Allowing the layers to separate

Fig. 32.10 Removing unreacted butan-1-ol

Step 5 Shake the crude 1-bromobutane with sodium hydrogencarbonate solution in another separating funnel. This removes the excess hydrochloric acid and other acidic impurities. The pressure due to the carbon dioxide produced must be released from time to time.

Repeat the washing until no effervescence occurs upon the addition of sodium hydrogencarbonate solution.

Allow the layers to separate and run the organic layer into a conical flask.

ASL 2009 Paper 1 Q7(e)
AL 2009 Paper 1 Q8(e)

The liquid is unclear before drying as tiny water droplets in 1-bromobutane cause the cloudiness.

Step 6 Add anhydrous sodium sulphate to dry the 1-bromobutane (Fig. 32.11). Swirl the mixture until the liquid is clear.

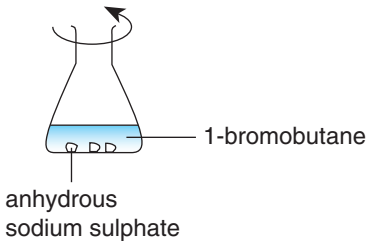


Fig. 32.11 Drying the product using anhydrous sodium sulphate

Collect the fraction that boils off at ±2 °C of the boiling point of the product.

Step 7 Decant the dried product into a pear-shaped flask. Carry out fractional distillation. Collect the fraction that boils off at 100–104 °C (the boiling point of 1-bromobutane is 102 °C). This removes any remaining organic impurities.

Find the mass of 1-bromobutane collected.

Calculating the percentage yield of the product

In a certain experiment, a student used 6.0 g of butan-1-ol with excess sodium bromide and concentrated sulphuric acid for the preparation. 6.5 g of 1-bromobutane were obtained.

$$\underset{74.0\ g\ mol^{-1}}{CH_3(CH_2)_2CH_2OH(aq)} + HBr(g) \longrightarrow \underset{136.9\ g\ mol^{-1}}{CH_3(CH_2)_2CH_2Br(l)} + H_2O(l)$$

According to the equation, 1 mole of C_4H_9OH reacts with excess hydrogen bromide to give 1 mole of C_4H_9Br.

Number of moles of C_4H_9OH reacted

$$= \frac{6.0\ g}{74.0\ g\ mol^{-1}}$$

$= 0.081$ mol

= number of moles of C_4H_9Br obtained theoretically

Theoretical yield of C_4H_9Br

$= 0.081\ mol \times 136.9\ g\ mol^{-1}$

$= 11$ g

Percentage yield of C_4H_9Br

$$= \frac{6.5\ g}{11\ g} \times 100\%$$

$= 59\%$

Most preparations give an actual yield that is less than the theoretical yield. There are several reasons why the yield may be low.

- The reaction may be incomplete — in this preparation, some unreacted butan-1-ol remains.
- There may be side-reactions occurring and these use up some of the reagents to create by-products — in this preparation, some of the butan-1-ol dehydrates to form an alkene when heated with concentrated sulphuric acid.
- Some of the product is lost during the preparation and purification processes.

32.1
Preparing and purifying 2-chloro-2-methylpropane.

Video
Preparing crude 2-chloro-2-methylpropane

Example 32.3

Q 2-chloro-2-methylpropane can be prepared by reacting methylpropan-2-ol with concentrated hydrochloric acid.

$(CH_3)_3COH(l) + HCl(aq) \longrightarrow (CH_3)_3CCl(l) + H_2O(l)$

This preparation follows the steps outlined below:

Step 1 Shake an excess amount of concentrated hydrochloric acid with methylpropan-2-ol in a separating funnel.

Step 2 Separate the organic layer from the aqueous layer.

Step 3 Wash the crude product with 10% sodium hydrogencarbonate solution.

DSE 2013 Paper 1A Q35

Step 4 Dry the product with a suitable reagent.

Step 5 Purify the dried product to remove the remaining methylpropan-2-ol.

The table below lists some information of methylpropan-2-ol and 2-chloro-2-methylpropane:

Compound	Methylpropan-2-ol	2-chloro-2-methylpropane
Density (g cm^{-3})	0.780	0.840
Boiling point (°C)	82.0	51.0
Water solubility	miscible	very slightly soluble
Molar mass (g mol^{-1})	74.0	92.5

Continued on next page

ASL 2011 Paper 1 Q9(b)–(d)
AL 2011 Paper 1 Q8(b)–(d)

a) Draw a diagram of the separating funnel after the reaction has taken place in *Step 1*, labelling clearly the aqueous layer and organic layer.

(The density of concentrated hydrochloric acid is 1.18 g cm^{-3}.)

b) Suggest ONE advantage of using a separating funnel to carry out *Step 1*.

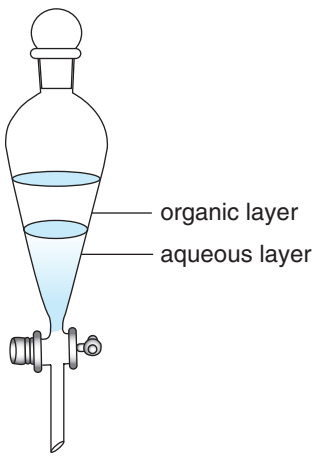
c) Outline the experimental procedure for washing the crude product in *Step 3*. Include the necessary safety precautions.

d) Suggest a suitable reagent for drying in *Step 4*.

e) Name a method for purifying the dried product in *Step 5*. — DSE 2013 Paper 1A Q35

f) Calculate the mass of product that would form from 5.00 cm^3 of methylpropan-2-ol if the percentage yield was 66.0%.

A1 a)



DSE 2013 Paper 1A Q35

b) Any one of the following:

- Allow better mixing of the reactants by vigorous shaking of the separating funnel.
- Allow easy isolation of the product by draining out the lower aqueous layer from the separating funnel.

c) Add slowly 10% sodium hydrogencarbonate solution. Wait until no effervescence occurs.

Stopper and shake the separating funnel vigorously.

During shaking, open the tap of the separating funnel regularly to release the pressure built inside the funnel.

Remove the aqueous layer. Repeat the washing procedure until no effervescence occurs upon the addition of sodium hydrogencarbonate solution.

d) Anhydrous calcium chloride / sodium sulphate

e) Simple distillation / fractional distillation

Continued on next page

f) Mass of methylpropan-2-ol used $= 5.00\ cm^3 \times 0.78\ g\ cm^{-3}$
$= 3.90\ g$

Number of moles of methylpropan-2-ol used
$= \dfrac{3.90\ g}{74.0\ g\ mol^{-1}}$
$= 0.0527\ mol$
= number of moles of product obtained theoretically

Theoretical yield of product $= 0.0527\ mol \times 92.5\ g\ mol^{-1}$
$= 4.87\ g$

Actual yield of product $= 4.87\ g \times 66.0\%$
$= 3.21\ g$

Practice 32.4

Ethanol reacts with ethanoic acid to produce ethyl ethanoate. The preparation of ethyl ethanoate consists of six steps as listed below:

Step 1	Heat 60.0 g of ethanol, 50.0 g of glacial ethanoic acid and some concentrated sulphuric acid under reflux for 10 minutes.
Step 2	Distil off the crude ester from the reaction mixture.
Step 3	Transfer the distillate to a separating funnel and shake with sodium carbonate solution.
Step 4	Remove the aqueous layer. Add some calcium chloride solution and shake. This removes the ethanol remaining as an impurity.
Step 5	Remove the aqueous layer. Transfer the organic layer to a conical flask. Add a few pieces of anhydrous calcium chloride.
Step 6	Decant the ester into a distillation flask. Carry out distillation. Collect the liquid that boils off at 74–79 °C.

e) i) Number of moles of CH_3CH_2OH used $= \dfrac{60.0\ g}{46.0\ g\ mol^{-1}}$
$= 1.30\ mol$

Number of moles of CH_3COOH used $= \dfrac{50.0\ g}{60.0\ g\ mol^{-1}}$
$= 0.833\ mol$

a) i) Write an equation for the reaction that occurs in *Step 1*. $CH_3CH_2OH(l) + CH_3COOH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$

ii) What is the function of concentrated sulphuric acid in the experiment? Concentrated sulphuric acid acts as a catalyst for the esterification reaction.

b) Explain why the crude ester is shaken with sodium carbonate solution in *Step 3*.
Refer to page T50.

c) Suggest how the aqueous layer can be removed in *Step 4*.
Refer to page T50.
(The density of ethyl ethanoate is 0.9 g cm^{-3}.)

d) What is the purpose of adding anhydrous calcium chloride in *Step 5*?
Refer to page T50.

e) In a certain experiment, 45.0 g of ethyl ethanoate were obtained.
ASL 2009 Paper 1 Q7(b)
AL 2009 Paper 1 Q8(b)

i) Calculate the number of moles of each reactant used.

ii) Calculate the percentage yield of ester in the experiment. 61.4%; refer to page T50.

(Molar masses of ethanol, ethanoic acid and ethyl ethanoate are 46.0 g mol^{-1}, 60.0 g mol^{-1} and 88.0 g mol^{-1} respectively.)

Worksheet **Assessment Form**
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Synthetic routes of important organic substances

Task

You are going to search for and present information about the synthetic routes of important organic substances commonly found in your daily lives, such as

- *aspartame*;
- *aspirin* (from salicylic acid); and
- *ibuprofen*.

Your teacher will divide the class into groups. Each group should design a poster to present the synthetic route of one of the substances named above. The groups may also choose other organic substances with the teacher's consent.

Reference websites

1 You may start your search from the following websites as appropriate:

- NutraSweet — Physical-chemical concepts — aspartame synthesis
  http://www.chemcases.com/nutra/nutra2.htm
- Aspirin
  http://www.ausetute.com.au/aspirin.html
- Ibuprofen
  http://www.rsc.org/learn-chemistry/resources/chemistry-in-your-cupboard/nurofen/4-6

2 Extend your search using search engines and appropriate keywords.

Refer to page T50.

aspartame 阿司百甜　aspirin 阿士匹靈　ibuprofen 布洛芬

Have you mastered?

Key terms

percentage yield 百分產率 136 overall yield 總百分產率 136 derivative 衍生物 137

distillation 蒸餾法 145 fractional distillation 分餾法 145 fractionating column 分餾柱 146

liquid-liquid extraction 液液萃取法 147 separating funnel 分液漏斗 147

re-crystallization 再結晶法 148

Checklist

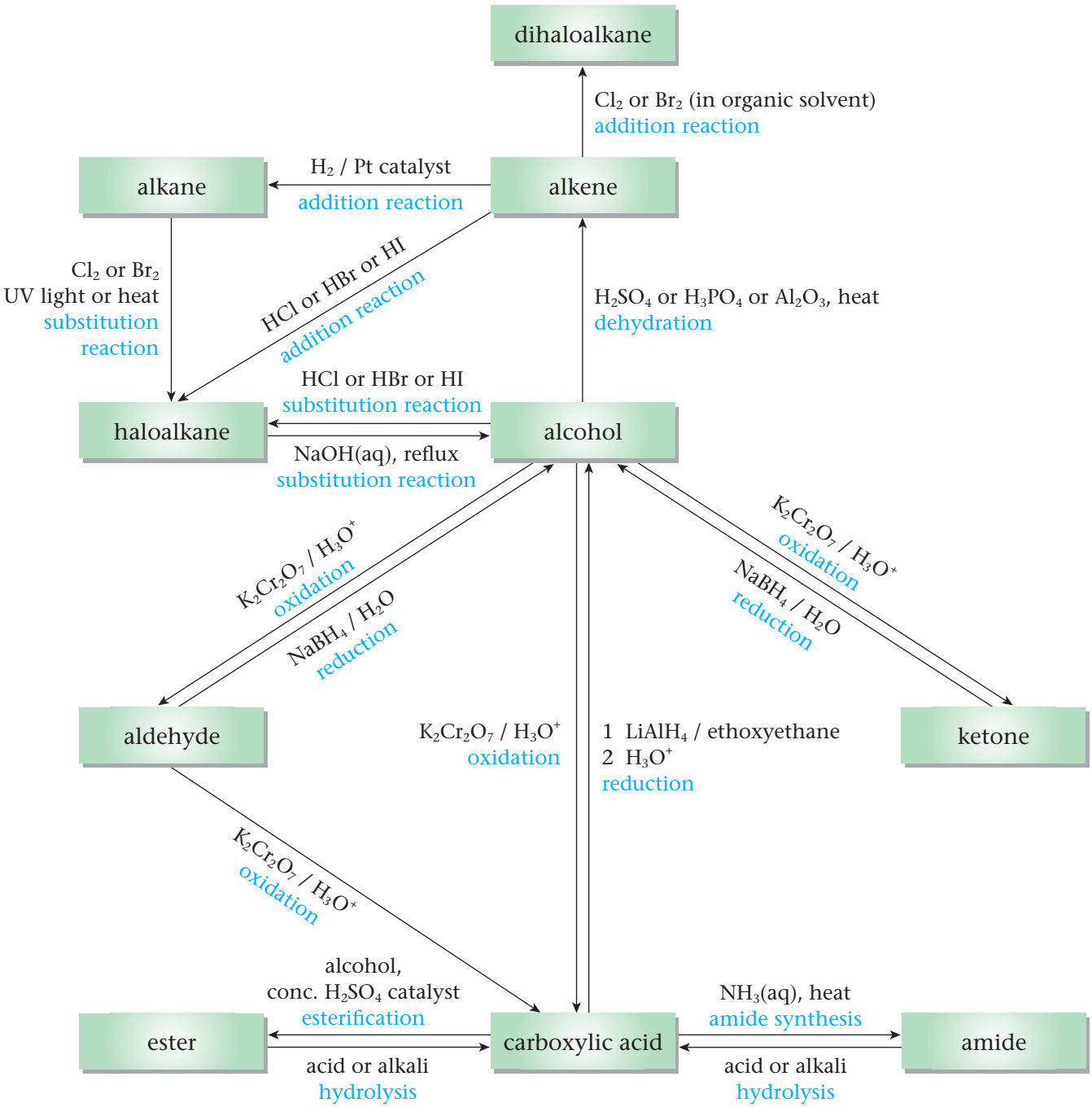
After studying this unit, you should be able to

- ☐ suggest routes to convert one functional group into another;
- ☐ state the reagents and conditions to accomplish conversions of carbon compounds;
- ☐ predict the major organic products of reactions;
- ☐ describe how to carry out laboratory preparation and purification of simple carbon compounds;
- ☐ calculate the percentage yield of a product obtained from a reaction.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The following flow chart summarizes important reactions of some homologous series.



Have you mastered?

2 When planning a synthesis, examine the target molecule. Then work backwards through a series of steps to find a suitable starting molecule.

3 The laboratory preparation of a carbon compound involves five key stages:

Stage 1 — planning;

Stage 2 — carrying out the reaction to produce the crude product;

Stage 3 — separating the crude product from the reaction mixture;

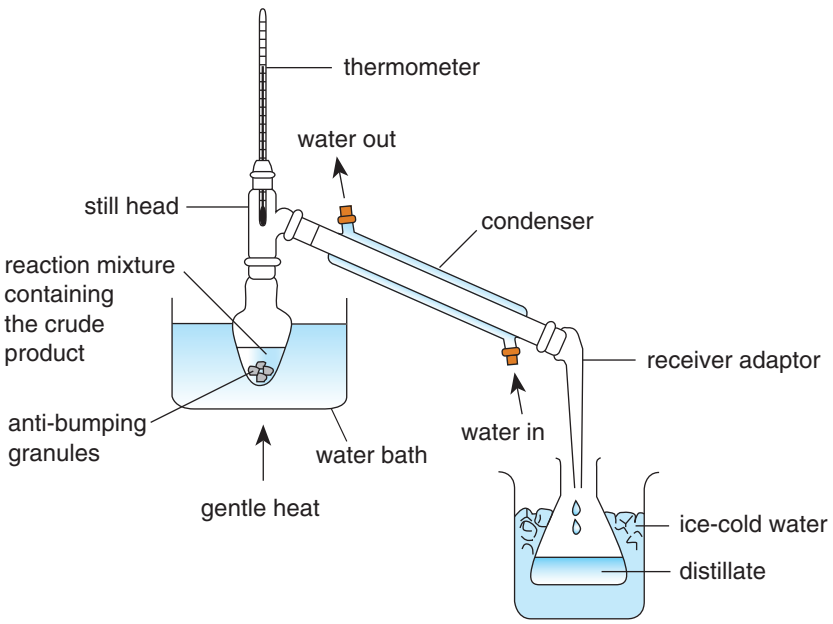
Stage 4 — purifying and drying the product;

Stage 5 — measuring the percentage yield of the product.

4 The table below summarizes the common separation and purification methods in carbon compound preparation.

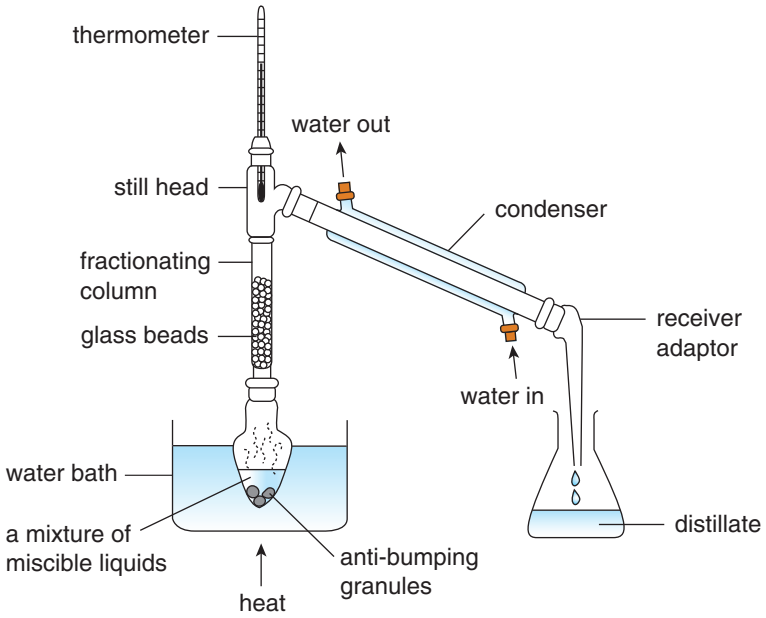
Common separation and purification methods	
Type of product	**Separation and purification method to employ**
Liquid product	• distillation • fractional distillation • liquid-liquid extraction
Solid product	• re-crystallization

a) Distillation

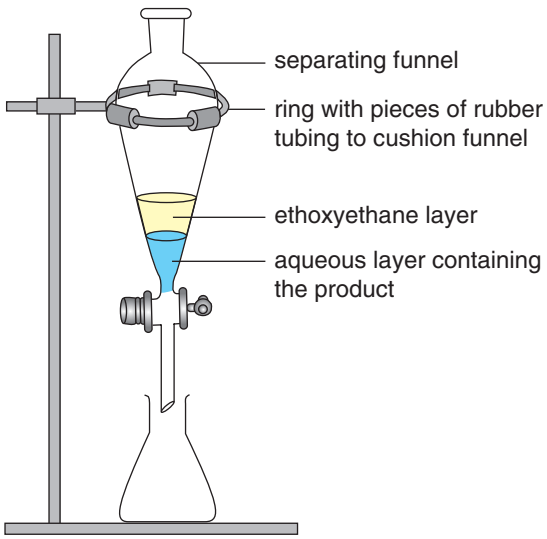


Have you mastered?

b) Fractional distillation



c) Liquid-liquid extraction

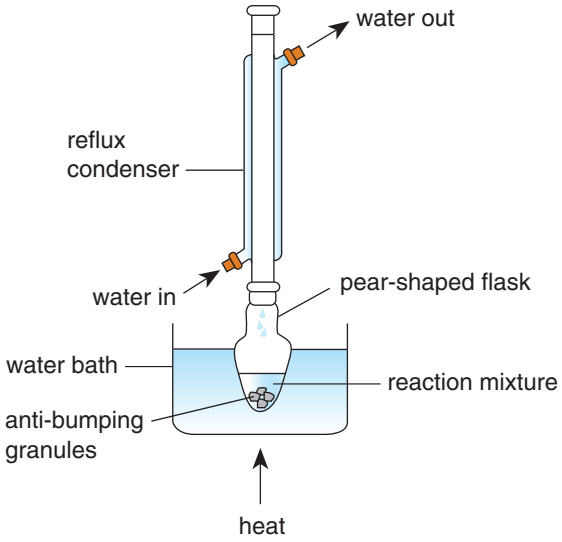


d) Re-crystallization

- Dissolve the crude solid product in the minimum amount of hot solvent and filter.
- Allow the filtrate to cool slowly for crystals to re-form. Filter off the crystals. Wash and dry them.

Have you mastered?

5 An experimental set-up for heating a reaction mixture under reflux:

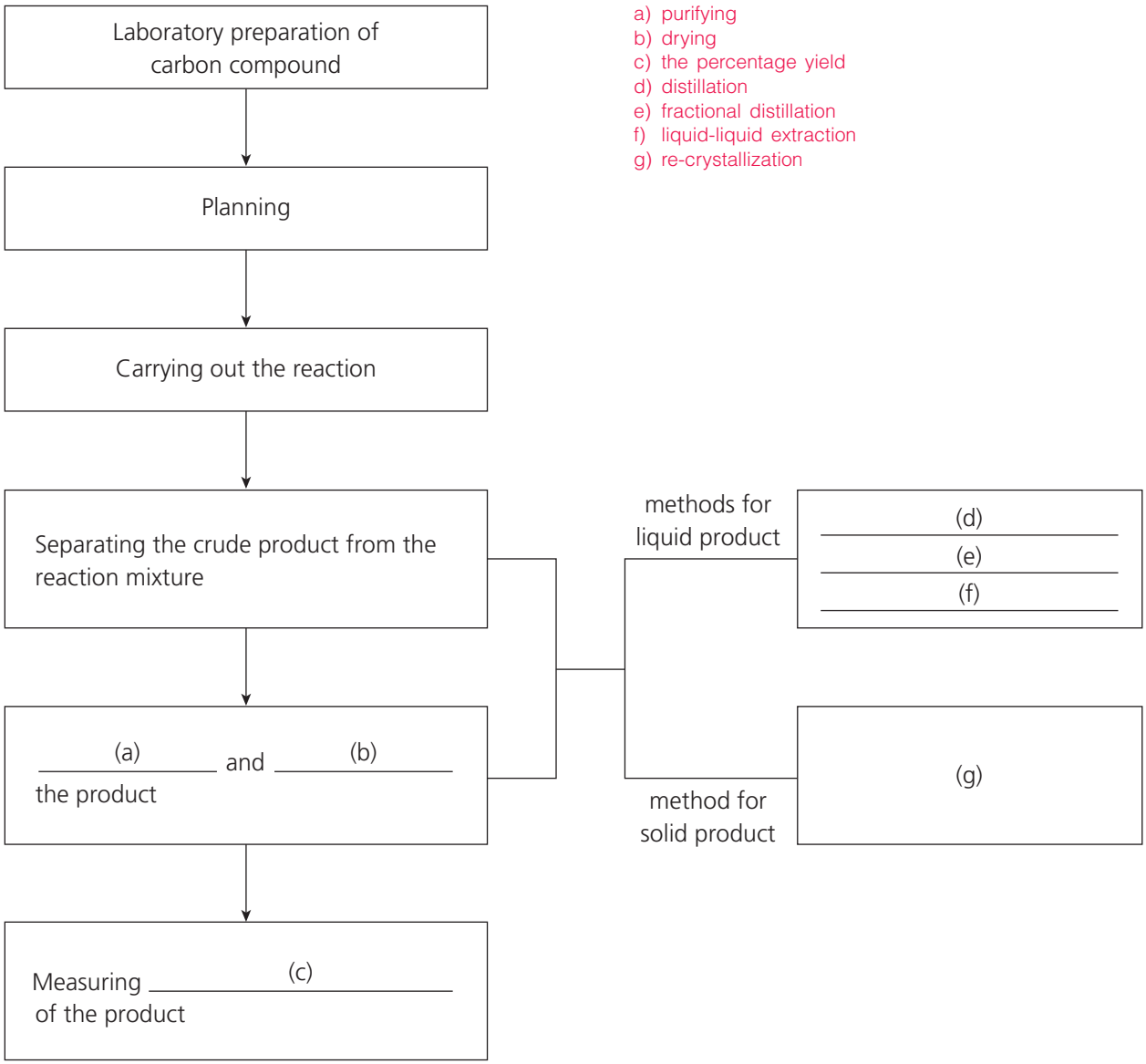


Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

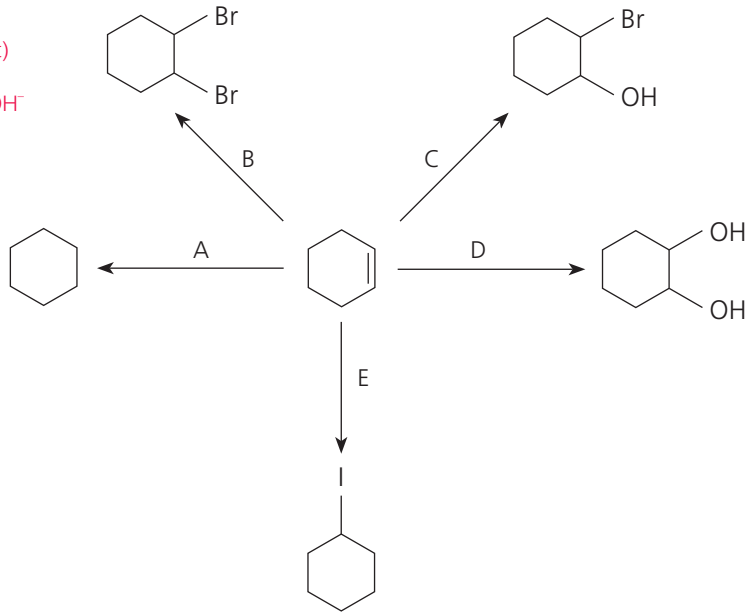
1 Complete the following concept map.



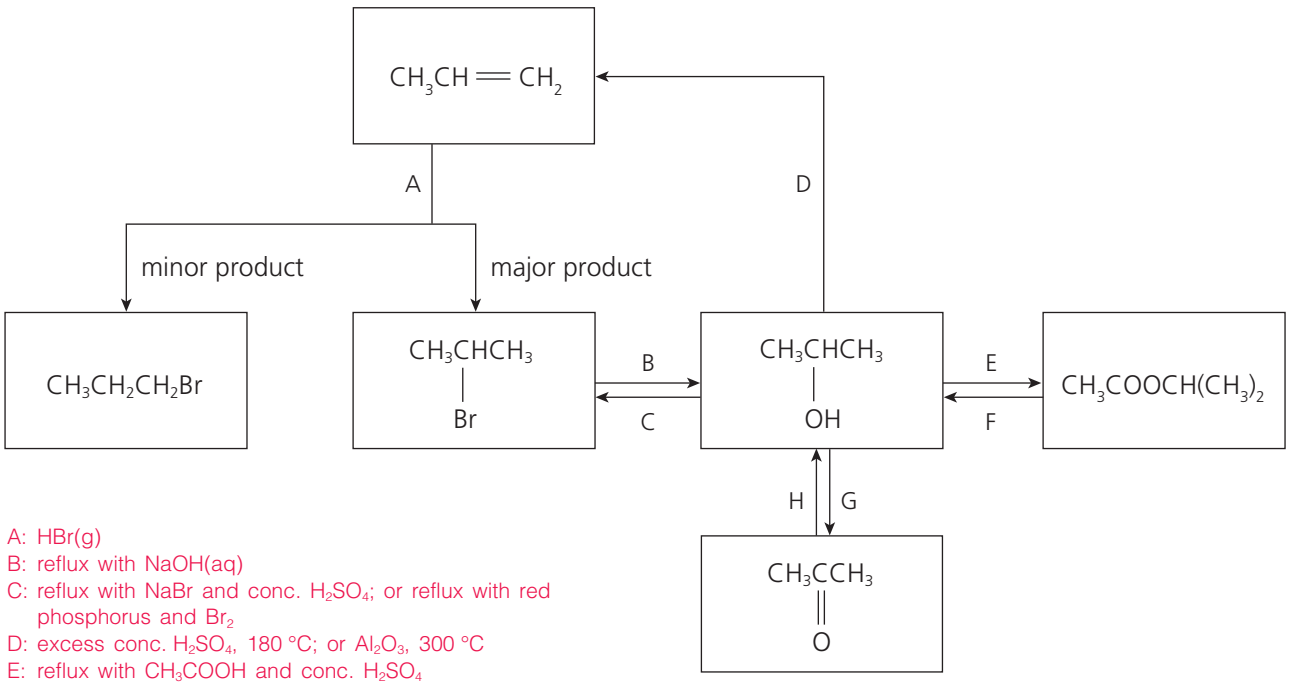
a) purifying
b) drying
c) the percentage yield
d) distillation
e) fractional distillation
f) liquid-liquid extraction
g) re-crystallization

2 Suggest the reagent(s) and reaction condition(s) for each of the following transformations from cyclohexene.

A: $H_2(g)$ / Pt catalyst
B: Br_2 (in organic solvent)
C: aqueous bromine
D: cold dilute $KMnO_4$ / OH^-
E: HI(g)

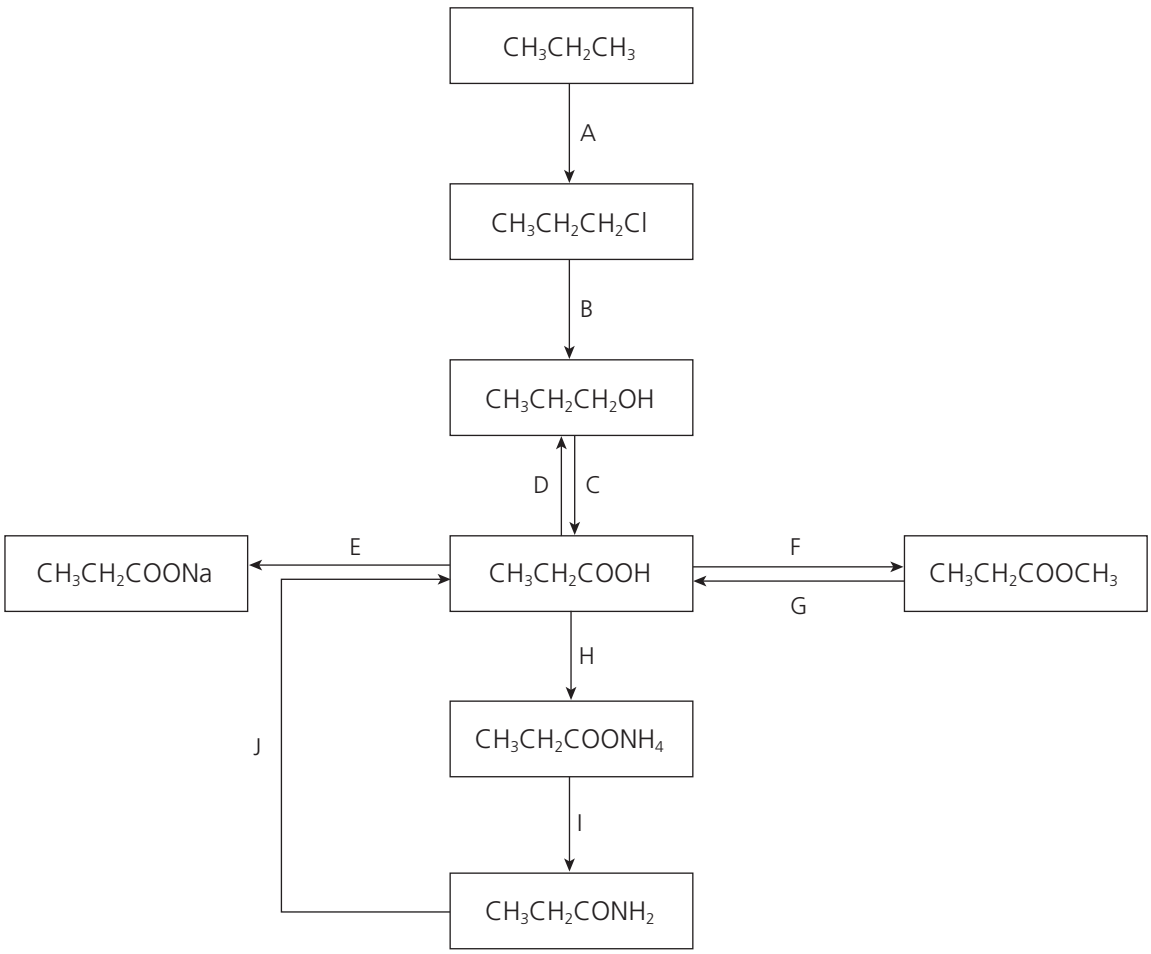


3 Suggest the reagent(s) and reaction condition(s) for each of the following transformations concerning propan-2-ol.



A: HBr(g)
B: reflux with NaOH(aq)
C: reflux with NaBr and conc. H_2SO_4; or reflux with red phosphorus and Br_2
D: excess conc. H_2SO_4, 180 °C; or Al_2O_3, 300 °C
E: reflux with CH_3COOH and conc. H_2SO_4
F: reflux with NaOH(aq)
G: $K_2Cr_2O_7$ / H_3O^+, heat
H: $NaBH_4$ / H_2O

4 Suggest the reagent(s) and reaction conditions(s) for each of the following transformations concerning propanoic acid.



A: $Cl_2(g)$, UV light
B: reflux with NaOH(aq)
C: $K_2Cr_2O_7$ / H_3O^+, reflux
D: 1 $LiAlH_4$ / ethoxyethane
2 H_3O^+
E: NaOH(aq)
F: reflux with CH_3OH and conc. H_2SO_4
G: reflux with NaOH(aq), separate the products by fractional distillation, obtain the acid by adding excess dilute sulphuric acid / hydrochloric acid to the sodium propanoate
H: $NH_3(aq)$
I: heat
J: reflux with NaOH(aq), obtain the acid by adding excess dilute sulphuric acid / hydrochloric acid to the sodium propanoate

Part II Multiple choice questions

5 In a laboratory experiment, a mixture of three alcohols (X, Y and Z) was separated into components by fractional distillation.

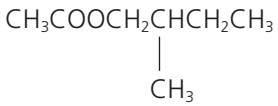
The first fraction collected is alcohol X, then alcohol Y followed by alcohol Z.

Which of the following can be deduced from the given information?

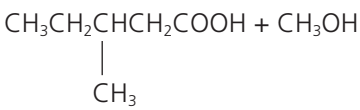
A Alcohol X has the highest boiling point among the three alcohols.
(B) Alcohol Y is more volatile than alcohol Z.
C The relative molecular mass of alcohol X is higher than that of alcohol Z.
D Alcohol Z has stronger covalent bonds in its molecules than alcohols X and Y.

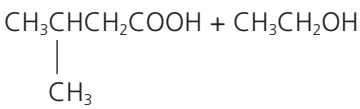
The order of the boiling points of the three alcohols is X < Y < Z. Thus, Y is more volatile than Z.

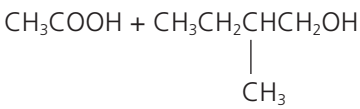
6 An ester with an odour of banana has the following structural formula:

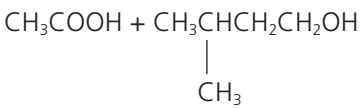


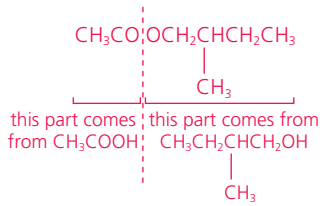
In which of the following pairs will the substances react together to produce this ester?

A 

B 

(C) 

D 



7 Which of the following reactions does NOT result in the product shown on the right-hand side of the reaction?

A $CH_3CH_2CH_2Cl \xrightarrow{NaOH(aq)} CH_3CH_2CH_2OH$

(B) $CH_3CH{=}CHCH_3 \xrightarrow{HCl(g)} CH_3CH_2CH_2CH_2Cl$

C $CH_3OH + CH_3COOH \xrightarrow{\text{conc. } H_2SO_4} CH_3COOCH_3$

D $CH_3CH{=}CH_2 \xrightarrow{Br_2(aq)} CH_3CH(OH)CH_2Br$

$CH_3CH{=}CHCH_3 \xrightarrow{HCl(g)} CH_3CH_2CHClCH_3$

8 Bromoethane can be made by heating ethanol under reflux with 50% sulphuric acid and sodium bromide. When the mixture is distilled, the products include sulphur dioxide, bromine, hydrogen bromide and water as well as bromoethane.

The product mixture is shaken with sodium carbonate solution and later with anhydrous sodium sulphate before being re-distilled. Which of the following shows the correct list of impurities removed at each step?

	Aqueous sodium carbonate wash	Addition of sodium sulphate
A	HBr	SO_2, Br_2, water
B	SO_2, Br_2	HBr, water
C	SO_2, HBr	Br_2, water
(D)	SO_2, Br_2, HBr	water

(Edexcel Advanced GCE, Unit 5, Feb. 2012, 16)

9 Consider the following organic conversion:

$(CH_3)_3COH \xrightarrow{X} (CH_3)_3CCl$

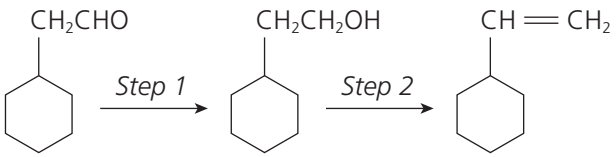
Which of the following reagents can X be?

(1) $Cl_2(g)$
(2) $PCl_3(l)$
(3) Concentrated HCl(aq)

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 33)

10 Consider the following conversion of organic compounds:



Which of the following statements about the above conversion are correct?

(1) $NaBH_4$ / H_2O can be used in *Step 1*.
(2) Excess concentrated H_2SO_4 can be used in *Step 2*.
(3) Heating is needed in *Step 2*.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

11 In order to prepare 2-chloro-2-methylpropane, a mixture of 2-methylpropan-2-ol and concentrated hydrochloric acid is shaken vigorously.

$$(CH_3)_3COH \xrightarrow{\text{conc. HCl}} (CH_3)_3CCl$$

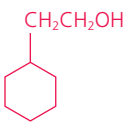
Which of the following statements about this preparation are correct?

(1) Two layers of liquids can be observed in the reaction mixture after shaking.
(2) The crude product should be washed with sodium carbonate solution.
(3) The unreacted 2-methylpropan-2-ol can be removed by simple distillation.

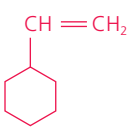
(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2013, 35)

10 (2) and (3) When heated with excess concentrated H_2SO_4,

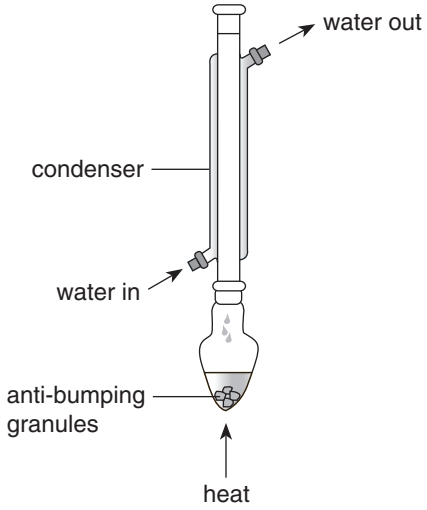


undergoes dehydration to produce



.

12 An experimental set-up is shown below.



Which of the following preparations can this set-up be used for?

(1) Preparing ethanal from ethanol, sodium dichromate solution and dilute sulphuric acid
(2) Preparing bromoethane from ethanol, sodium bromide and concentrated sulphuric acid
(3) Preparing 1,2-dibromoethane from ethene and bromine

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1) Heating ethanol, sodium dichromate solution and dilute sulphuric acid under reflux produces ethanoic acid, NOT ethanal.
(3) Ethene and bromine react quickly at room temperature. There is NO need to heat under reflux.

Unit Exercise

Part III Structured questions

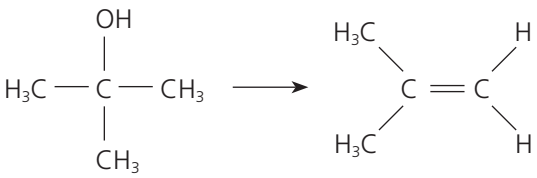
Answers for the questions of the public examinations in Hong Kong are not provided.

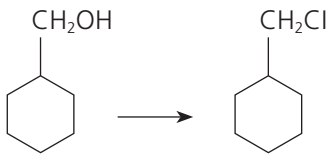
13 Suggest the reagent(s) and reaction condition(s) for each of the following conversions. Name also the type of reaction that takes place in each case.

a) $CH_3CH{=}CH_2 \longrightarrow CH_3CHBrCH_3$

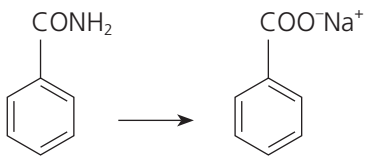
b) $CH_3CH_2CH_3 \longrightarrow CH_3CHBrCH_3$

c) $CH_3CHBrCH_3 \longrightarrow CH_3CH(OH)CH_3$

d) 

e) 

f) $CH_2{=}CHCH_2CHO \longrightarrow CH_2{=}CHCH_2CH_2OH$

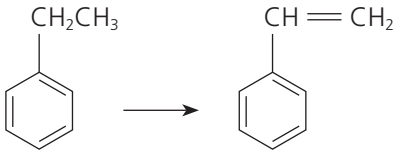
g) 

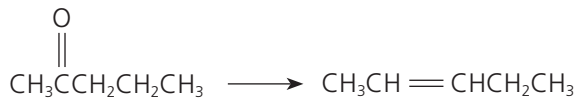
h) $CH_3CH_2COOCH_2CH_3 \longrightarrow CH_3CH_2COOH$

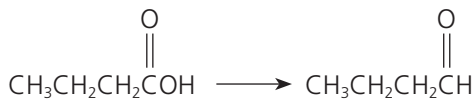
a) Reagent(s) and reaction condition(s): HBr(g)
Type of reaction: addition
b) Reagent(s) and reaction condition(s): expose a mixture of $CH_3CH_2CH_3$ and Br_2 to ultraviolet light or heat
Type of reaction: substitution
c) Reagent(s) and reaction condition(s): reflux with NaOH(aq)
Type of reaction: hydrolysis / substitution
d) Reagent(s) and reaction condition(s): 20% H_2SO_4, heat
Type of reaction: dehydration / elimination
e) Reagent(s) and reaction condition(s): reflux with conc. HCl + $ZnCl_2$ catalyst; or mix with PCl_5; or reflux with $SOCl_2$
Type of reaction: substitution
f) Reagent(s) and reaction condition(s):
1 $LiAlH_4$ / ethoxyethane
2 H_3O^+; or
$NaBH_4$ / H_2O
Type of reaction: reduction
g) Reagent(s) and reaction condition(s): NaOH(aq), heat
Type of reaction: hydrolysis
h) Reagent(s) and reaction condition(s):
1 reflux with NaOH(aq)
2 separate the products by fractional distillation
3 obtain the acid by adding excess dilute sulphuric acid / hydrochloric acid to the sodium propanoate
Type of reaction: hydrolysis

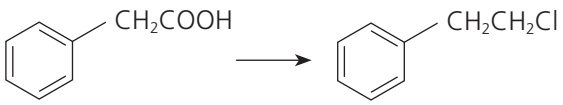
14 Outline a synthetic route, in NOT MORE THAN FOUR STEPS, to accomplish each of the following conversions. For each step, give the reagent(s), the condition(s) and the structure(s) of the intermediate product(s).

Refer to page T55.

a) 

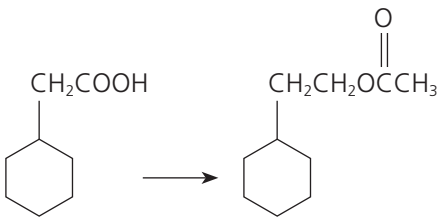
b) 

c) 

d) 

e) $CH_2{=}CHCl \longrightarrow CH_3CHO$

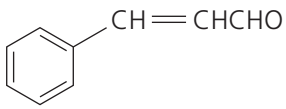
f) $CH_3CH(OH)CH_3 \longrightarrow CH_3CHCH_2$ (OH, OH)

g) 

h) $C_2H_5OH \longrightarrow CH_3CONH_2$

Unit Exercise

15 Cinnamon, which can be used as a flavouring, contains cinnamaldehyde (C_9H_8O). The structure of cinnamaldehyde is shown below:



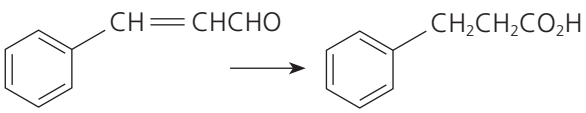
a) Draw the *trans*-isomer for the above structure.

b) Explain why ethyl ethanoate is a better solvent than water for dissolving cinnamaldehyde.

c) In an experiment to extract cinnamaldehyde from cinnamon, a solution containing only ethyl ethanoate and cinnamaldehyde is obtained after a series of steps. In order to separate these two compounds, simple distillation can be carried out. Draw a diagram for the set-up involved, and label the name of the distillate collected.

(Boiling points: cinnamaldehyde = 248 °C, ethyl ethanoate = 77 °C)

d) Outline a synthetic route, with no more than three steps, to accomplish the following conversion. For each step, give the reagent(s), reaction conditions (as appropriate) and structure of the organic product.



(HKDSE, Paper 1B, 2012, 12(a)–(d))

16 An experiment to prepare 0.100 mole of 1-bromobutane uses the reaction of butan-1-ol with hydrogen bromide.

Refer to page T56.

Hydrogen bromide is formed in the reaction mixture from potassium bromide and moderately concentrated sulphuric acid.

The process has an 80% yield after purification of the 1-bromobutane.

$$KBr + H_2SO_4 \longrightarrow KHSO_4 + HBr$$

$$CH_3CH_2CH_2CH_2OH + HBr \longrightarrow CH_3CH_2CH_2CH_2Br + H_2O$$

The steps of the experimental procedure are as follows.

1 Add measured amounts of potassium bromide and butan-1-ol to 10 cm^3 of water into a 50 cm^3 two-necked flask.

2 Fit the two-necked flask with a reflux condenser and a tap funnel.

3 Immerse the flask in a beaker of cold water and add 10 cm^3 of concentrated sulphuric acid from the tap funnel, a few drops at a time.

4 Remove the flask from the cold water and close the tap on the tap funnel. Heat the mixture under reflux for 30 minutes.

5 Allow the mixture to cool. Then set up the apparatus for distillation. Boil the mixture and collect the distillate in a measuring cylinder.

6 Transfer the distillate to a separating funnel. The distillate consists of two layers, an aqueous layer and impure 1-bromobutane. Separate the two layers.

7 Wash the impure 1-bromobutane with concentrated hydrochloric acid and separate the two layers.

8 Wash the 1-bromobutane layer with sodium hydrogencarbonate solution, releasing any gas formed.

9 Collect the 1-bromobutane layer in a conical flask and add anhydrous sodium sulphate.

10 Decant the 1-bromobutane into a 50 m^3 flask.

Data

Property	Butan-1-ol	1-bromobutane	Water
Density ($g\ cm^{-3}$)	0.81	1.3	1.0
Molar mass ($g\ mol^{-1}$)	74.0	137.0	18.0
Boiling temperature (°C)	117.3	101.7	100.0

a) i) Show, by calculation, that 0.125 mole of butan-1-ol is needed to make 0.100 mole of 1-bromobutane.

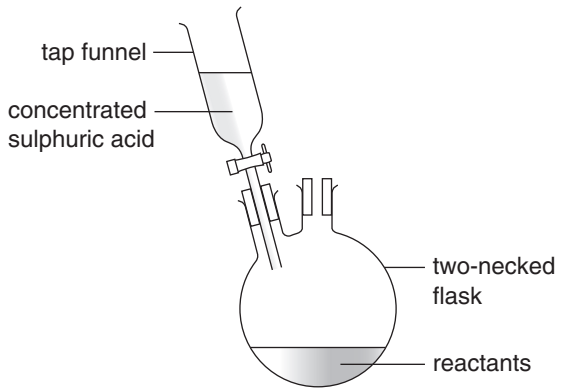
ii) Calculate the volume of 0.125 mole of butan-1-ol, in cm^3. 11.4 cm^3

iii) Calculate the minimum mass of potassium bromide required in *Step 1*. 14.9 g

(The molar mass of potassium bromide is 119.0 g mol^{-1})

b) Complete and label the diagram below of the apparatus assembled in *Steps 1*, *2* and *3*.

(You may assume that the apparatus is suitably clamped)



c) i) State, with a reason, whether the upper or lower layer contains 1-bromobutane in *Step 6*.

ii) The product is washed with concentrated hydrochloric acid in *Step 7* to remove unreacted butan-1-ol. In *Step 8*, why is the product then washed with sodium hydrogencarbonate solution and what causes a build up of gas?

d) i) What further step is necessary to purify the 1-bromobutane obtained in *Step 10*?

ii) How does the step in (d)(i) give information about the purity of the product?

(Edexcel Advanced Subsidiary GCE, Unit 3B, May 2010, 4)

17 2-chloro-2-methylpropane is prepared by shaking 15 cm^3 of 2-methylpropan-2-ol with 40 cm^3 of concentrated hydrochloric acid (excess) in a 100 cm^3 separating funnel.

$$(CH_3)_3COH \xrightarrow{\text{conc. HCl(aq)}} (CH_3)_3CCl$$

The table below lists some information of 2-methylpropan-2-ol and 2-chloro-2-methylpropane:

Compound	2-methylpropan-2-ol	2-chloro-2-methylpropane
Relative molecular mass	74.0	92.5
Density (g cm^{-3})	0.78	0.84
Boiling point (°C)	82	51
Solubility in water	miscible	very slightly soluble

a) What would be observed in the separating funnel after the reaction has taken place?

b) Suggest ONE advantage of using a separating funnel to carry out the experiment.

c) The crude product of 2-chloro-2-methylpropane is washed with 10% $NaHCO_3$(aq). Outline the experimental procedures involved, including the necessary safety precautions.

d) Suggest a suitable reagent for drying the washed product from (c).

e) Draw a labelled diagram of the assembled apparatus for the purification of the dried product from (d) to remove the remaining 2-methylpropan-2-ol.

f) After purification, 5.8 g of 2-chloro-2-methylpropane is obtained. Calculate the percentage yield of the preparation.

(HKASLE, Paper 1, 2011, 9)

18 Propanoic acid may be prepared by oxidizing propan-1-ol in acidic conditions.

$$CH_3CH_2CH_2OH \xrightarrow{X + H_2SO_4} CH_3CH_2COOH$$

Procedure

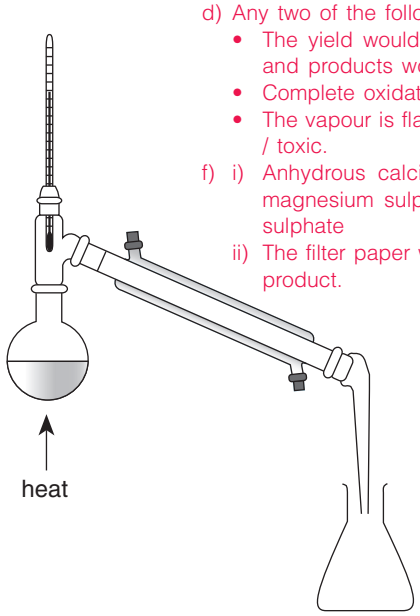
1 Pour 10 cm^3 of distilled water into a boiling tube and add 12 g of oxidizing agent X. Shake the mixture and leave X to dissolve.

2 Pour 3 cm^3 of propan-1-ol into a round-bottom flask and add 10 cm^3 of distilled water and a few anti-bumping granules. Set up the apparatus for heating under reflux.

3 Add 4 cm^3 of concentrated sulphuric acid, drop by drop, to the propan-1-ol. While the mixture is still warm, add the solution of oxidizing agent X, drop by drop. The energy released from the reaction should cause the mixture to boil without external heating.

4 When all of the solution of X has been added, use a low Bunsen burner flame to keep the mixture boiling for 10 minutes, not allowing any vapour to escape.

5 Distil the mixture in the flask using the apparatus shown below. Collect 5–6 cm^3 of distillate, which is an aqueous solution of propanoic acid.



a) Suggest, a suitable oxidizing agent, X, for this reaction.
Sodium / potassium dichromate solution

b) What colour change does X undergo when it oxidizes propan-1-ol?
From orange to green

c) Draw a labelled diagram showing the apparatus for heating under reflux.
Refer to page T57.

d) Give TWO reasons why the escape of vapour in *Step 4* should be prevented.

e) How does the reflux apparatus prevent the escape of vapour?
The vapour is cooled and condensed.

f) Some water can be removed from the distillate in *Step 5* by adding a solid drying agent. The solution of propanoic acid can then be decanted, leaving the drying agent behind.

i) Suggest a suitable solid drying agent.

ii) Suggest why removing excess solid drying agent by decanting, rather than filtering through filter paper, improves the yield.

g) In a larger-scale preparation of propanoic acid, 10.0 g of propan-1-ol was used.
Refer to page T57.

i) Calculate the maximum mass of propanoic acid which could be formed from 10.0 g of propan-1-ol. 12.4 g

Compound	Propan-1-ol	Propanoic acid
Molar mass ($g\ mol^{-1}$)	60.0	74.0

ii) After purification, 6.0 cm^3 of dry propanoic acid was obtained.

Calculate the percentage yield in the preparation. 47.9%

The density of propanoic acid is 0.99 $g\ cm^{-3}$.

h) In another experiment, the same reaction mixture (propan-1-ol, X and concentrated sulphuric acid) was heated in the apparatus shown in *Step 5*. Identify the main organic product which would be collected and explain why propanoic acid is not produced.

(Edexcel Advanced Subsidiary GCE, Unit 3B, Jan. 2012, 5)

d) Any two of the following:
- The yield would be reduced. / Reactants and products would be lost.
- Complete oxidation could not occur.
- The vapour is flammable / harmful / irritant / toxic.

f) i) Anhydrous calcium chloride / anhydrous magnesium sulphate / anhydrous sodium sulphate
ii) The filter paper would absorb some of the product.

CH_3CH_2CHO
Only partial oxidation of propan-1-ol occurs / incomplete oxidation of propan-1-ol / product removed as formed.

19 The following paragraph was extracted from the laboratory report of a student on the hydrolysis of ethyl benzoate.

'A mixture of ethyl benzoate and excess sodium hydroxide solution was placed in a flask and heated under reflux until the hydrolysis was complete. The reaction mixture was then distilled and a distillate was collected at about 78 °C. The solution remaining in the flask was allowed to cool.'

a) Draw a labelled diagram for the laboratory set-up used for heating ethyl benzoate and sodium hydroxide solution under reflux.

b) Write an equation for the hydrolysis.

c) Give the systematic name of the main constituent in the distillate.

d) Suggest how a crude sample of benzoic acid can be obtained.

e) The crude sample of benzoic acid can be purified by re-crystallization from hot water. Outline the procedure in the re-crystallization process.

f) In an experiment, 3.30 g of ethyl benzoate gave 1.90 g of benzoic acid. Calculate the percentage yield of benzoic acid.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0)

20 State a possible consequence from each of the following poor laboratory techniques:

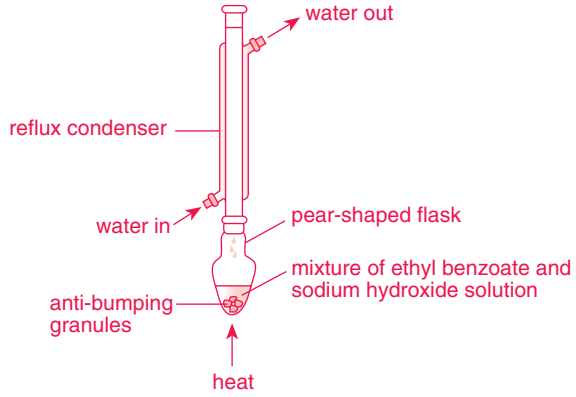
a) shaking a liquid mixture inside a separating funnel without opening the tap regularly;

b) running off the lower layer from a separating funnel without removing the stopper;

c) using an excess of a hot solvent to dissolve the crude solid product during re-crystallization.

a) Shaking and the heat of the hands may cause the liquid inside the separating funnel to vaporize. Pressure may build up inside the funnel.
An explosion may occur if the tap is not opened regularly for pressure release.
b) Without releasing the pressure, the liquid in the separating funnel will not drain out of the funnel.
c) The product may not separate out as crystals if too much solvent is used.

19 a)



b) $C_6H_5COOCH_2CH_3$(l) + NaOH(aq) → $C_6H_5COO^-Na^+$(aq) + CH_3CH_2OH(aq)

c) ethanol
d) Add dilute sulphuric acid / hydrochloric acid to the solution remaining in the flask.
Then filter.
e) Dissolve the crude sample in the minimum amount of hot water.
Filter to remove the insoluble impurities.
Allow the filtrate to cool slowly for crystals to re-form.
Filter off the crystals.
Wash and dry.
f) Number of moles of ethyl benzoate = $\frac{3.30\text{ g}}{150.0\text{ g mol}^{-1}}$
= 0.0220 mol
According to the equation, 1 mole of $C_6H_5COOCH_2CH_3$ gives 1 mole of C_6H_5COOH upon hydrolysis.
i.e. number of moles of C_6H_5COOH = 0.0220 mol
Theoretical yield of C_6H_5COOH = 0.0220 mol x 122.0 g mol^{-1}
= 2.68 g
Percentage yield of C_6H_5COOH = $\frac{1.90\text{ g}}{2.68\text{ g}}$ x 100%
= 70.9%